

CEFATUM INJECTION

DESCRIPTION:

CEFATUM INJECTION: White to cream coloured crystalline powder.

COMPOSITION:

CEFATUM INJECTION 1G: Each vial of 10 ml contains Ceftazidime (as Ceftazidime Pentahydrate with Sodium Carbonate) 1 g

CEFATUM INJECTION 2G: Each vial of 20 ml contains Ceftazidime (as Ceftazidime Pentahydrate with Sodium Carbonate) 2 g

PHARMACODYNAMICS:

Mechanism of action / Effect: Bactericidal; action depends on ability to reach and bind penicillin-binding proteins located in bacterial cytoplasmic membranes; cephalosporins inhibit bacterial septum and cell wall synthesis, probably by acylation of membrane-bound transpeptidase enzymes. This prevents cross-linkage of peptidoglycan chains, which is necessary for bacterial cell wall strength and rigidity. Also, cell division and growth are inhibited, and lysis and elongation of susceptible bacteria frequently occur. Rapidly dividing bacteria are those most susceptible to the action of cephalosporins.

Ceftazidime is bactericidal in action, exerting its effect on target cell wall proteins and causing inhibition of cell wall synthesis. It is stable to most beta-lactamases produced by Gram-positive and Gram-negative organisms and consequently is active against many ampicillin- and cephalothin-resistant strains (but not methicillin-resistant strains). Ceftazidime has been shown to have *in vitro* activity against the following organisms:

Gram negative: *Pseudomonas aeruginosa*, *Pseudomonas* species (other), *Klebsiella pneumoniae*, *Klebsiella* species (other), *Proteus mirabilis*, *Proteus vulgaris*, *Morganella morganii* (formerly *Proteus morganii*), *Proteus rettgeri*, *Providencia* species, *Escherichia coli*, *Enterobacter* species, *Citrobacter* species, *Serratia* species, *Acinetobacter* species, *Neisseria gonorrhoea*, *Neisseria meningitidis*, *Haemophilus influenzae* (including ampicillin-resistant strains).

Gram positive: *Staphylococcus aureus* (methicillin-sensitive strains), *Staphylococcus epidermidis* (methicillin-sensitive strains), *Micrococcus* species, *Streptococcus pyogenes*, *Streptococcus* Group B, *Streptococcus pneumoniae*, *Streptococcus* species (excluding *Streptococcus faecalis*).

Ceftazidime is not active *in vitro* against methicillin-resistant staphylococci, *Streptococcus faecalis* and many other *Enterococci*, *Listeria monocytogenes*, *Campylobacter* species or *Clostridium difficile*.

In vitro, the activities of ceftazidime and aminoglycoside antibiotics in combination have been shown to be at least additive; there is evidence of synergy in some strains tested. This property may be important in the treatment of febrile neutropenic patients.

PHARMACOKINETICS:

Absorption of ceftazidime after oral administration is negligible, therefore ceftazidime is intended for parenteral use only. In man after a single intramuscular administration of 500 mg and 1 g, mean peak serum levels of 18 and 37 mg/L respectively are achieved at 1 hour falling to 8 and 2 mg/L and 20 and 5 mg/L at 4 and 8 hours respectively for two doses. Five minutes after an intravenous bolus injection of 500 mg, 1 g and 2 g, mean serum levels are respectively 46, 87 and 170 mg/L falling to 17 and 6 mg/L, 32 and 10 mg/L and 85 and 15 mg/L at 1 and 4 hours respectively with the three doses. The serum half-life in adults with normal renal function is about 1.8 hours (1.2-2.9 hours). This may be prolonged to 20-35 hours in anuric patients. In neonates, the serum half-life of ceftazidime can be 3-4 times greater than that measured in adults. The serum protein binding of ceftazidime is low at about 10%.

Ceftazidime is not metabolised in the body and is excreted unchanged in the active form into the urine by glomerular filtration. In the presence of normal renal function approximately 80 to 90% of the dose is recovered in the urine within 24 hours. Less than 1% is excreted via the bile.

The mean maximum concentrations of ceftazidime in bone, heart, bile, sputum, aqueous humour, synovial and pleural and peritoneal fluids were in excess of the *in vitro* minimum inhibitory levels for susceptible organisms. Transplacental transfer of the antibiotic readily occurs. Ceftazidime penetrates the intact blood brain barrier poorly and low levels are achieved in the CSF.

The pharmacokinetics of ceftazidime are similar whether it is administered by a single or by repeat dosage.

Concurrent oral administration of probenecid did not affect the serum levels or urinary recoveries of ceftazidime. The pharmacokinetics of ceftazidime were not affected when administered intramuscularly with 0.5% lignocaine.

INDICATION:

Ceftazidime is indicated for the treatment of single and mixed infections caused by susceptible aerobic organisms with suspected resistance to other antimicrobials, but not to ceftazidime, and as an alternative to aminoglycosides in pseudomonal infection in patients in whom aminoglycosides toxicity is a cause for concern and other pseudomonal antibiotics cannot be used.

Indications include:

Septicaemia including neonatal sepsis.

Bacteraemia eg. infected burns.

Respiratory infections: pneumonia, broncho- pneumonia, infected pleurisy, infected bronchiectasis and bronchitis.

Severe ear, nose and throat infections: otitis media, mastoiditis.

Urinary tract infections: acute and chronic pyelonephritis, pyelitis, cystitis, urethritis (bacterial only) and infections associated with bladder and renal stones. Skin and soft tissue infections: erysipelas, abscesses, cellulitis, infected burns and wounds, mastitis.

Gastrointestinal and abdominal infections: intra-abdominal abscesses, enterocolitis.

Bone and joint infections: osteitis, osteomyelitis, septic arthritis, infected bursitis.

RECOMMENDED DOSAGE:

Dosage and administration:

General dosage recommendations: Ceftazidime is to be used by the parenteral route, the dosage depending upon the severity, sensitivity and type of infection and the age, weight and renal function of the patient.

Adults: The adult dosage range for is 1 to 6 g per day: for instance, 500 mg, 1 g or 2 g given 12 or 8 hourly by i.v. or i.m. injection. In urinary tract infections and in many less serious infections, 500 mg or 1 g 12 hourly is usually adequate. In the majority of infections, 1 g 8 hourly or 2 g 12 hourly should be given. In very severe infections, 2 g 8 or 12 hourly should be administered. Individual doses in excess of 1 g should be administered intravenously.

Infants and children: The usual dosage range for children aged over 12 months is 25 to 100 mg/kg/day (up to a maximum of 6 g/day), given as two or three divided doses. The maximum daily dosage (6 g) may be given to children with very serious infections eg. those who are immunocompromised or who suffer from cystic fibrosis.

Neonates and infants up to 12 months: 25-100 mg/kg/day in two divided doses. In neonates the serum half-life of ceftazidime can be 3-4 times greater than that measured in adults.

Use in the elderly: In view of the reduced clearance of ceftazidime in elderly patients, the daily dose should be adjusted according to renal function.

Dosage in impaired renal function: Ceftazidime is excreted by the kidneys almost exclusively by glomerular filtration. Therefore, in patients with impaired renal function it is recommended that the dosage of ceftazidime should be reduced to compensate for its slower excretion, except in mild impairment, ie. glomerular filtration rate (GFR) greater than 50 mL/min. In patients with suspected renal insufficiency, an initial loading dose of 1 g ceftazidime may be given. An estimate of GFR should be made to determine the appropriate maintenance dose. Recommended maintenance doses are shown below:

Recommended maintenance doses of ceftazidime in renal insufficiency

Creatinine clearance mL/min	Approx. serum creatinine* micromole/L	Recommended Unit dose of ceftazidime g	Frequency of dosing Hourly
50-31	150-200	1.0	12
30-16	200-350	1.0	24
15-6	350-500	0.5	24
5	500	0.5	48

*These values are guidelines and may not accurately predict renal function in all patients especially in the elderly in who the serum creatinine concentration may overestimate renal function.

In patients with severe infections who would normally receive 6 g of ceftazidime daily were if not for renal insufficiency, the unit dose given in the table above may be increased by 50% or the dosing frequency increased appropriately. In such patients it is recommended that ceftazidime serum levels should be monitored and trough levels should not exceed 40 mg/L.

When only serum creatinine is available, the following formula (Cockcroft's equation) may be used to estimate creatinine clearance. The serum creatinine should represent a steady state of renal function:

Males: Creatinine clearance (mL/min) = $\frac{\text{Weight (kg)} \times (140 - \text{age in years}) \times 88.4}{72 \times \text{serum creatinine (micromole/L)}}$

Females: 0.85 x above value.

In children the creatinine clearance should be adjusted for body surface area or lean body mass and the dosing frequency reduced in cases of renal insufficiency as for adults.

The serum half-life of ceftazidime during haemodialysis is approximately 3 hours. The appropriate maintenance dose of ceftazidime should be repeated following each haemodialysis period. Continuous ambulatory peritoneal dialysis (CAPD) removed approximately 10% of the antibiotic when the dwell time was 4-6 hours.

Administration: Ceftazidime may be given intravenously or by deep intramuscular injection into a large muscle mass such as the upper outer quadrant of the gluteus maximus or lateral part of the thigh.

Instructions for reconstitution: Ceftazidime may be constituted with water for injection or, for intramuscular injection, with 0.5% lignocaine. See table for addition volumes and solution concentrations.

Table: Preparation of Solution

Vial Size	Amount of Diluent to be added	Approximate Concentration (mg/mL)
1g		
- intramuscular	3.0 mL	260
- intravenous	10 mL	90
2g		
- intravenous	10 mL	170

For ease of use, it is recommended that the following techniques of reconstitution are adopted.

1 g i.m./i.v. and 2 g i.v. bolus vials:

Insert the syringe needle through the vial closure and inject the recommended volume of diluent. Remove the syringe needle. Shake to dissolve: carbon dioxide is released and a clear solution obtained in about 1-2 minutes. Invert the vial. With the syringe plunger fully depressed, insert the needle through the vial closure and withdraw the total volume of solution into the syringe. Ensure that the needle remains within the solution and does not enter the headspace. The withdrawn solution may contain small bubbles of carbon dioxide; they may be disregarded.

These solutions may be given directly into the vein or introduced into the tubing of a giving set if the patient is receiving parenteral fluids.

Solutions of ceftazidime injection reconstituted in water for injections retain satisfactory potency for 12 hours if kept below 25°C or for 7 days if refrigerated (2- 8°C). When reconstituted in 0.5% Lignocaine Hydrochloride injection BP the corresponding times are 6 hours at below 25°C or 4 days under refrigeration (2-8°C). Some increase in the colour of prepared solutions of ceftazidime for injection may occur on storage. It is, however, advisable to use the reconstituted product as soon as possible.

Ceftazidime is compatible with the intravenous fluids shown below. Solutions at concentrations between 1 mg/mL and 40 mg/mL in these infusion fluids may be stored for up to 12 hours below 25°C or 7 days if refrigerated (2-8°C).

0.9% Sodium Chloride Injection BP

M/6 Sodium Lactate Injection BP

M/6 Compound Sodium Lactate Injection BP (Hartmann's Solution)

5% Glucose Injection BP

Dextran 40 Injection BP 10% in 0.9% Sodium Chloride Injection BP

Dextran 40 Injection BP 10% in 5% Glucose Injection BP

Dextran 70 Injection BP 6% in 0.9% Sodium Chloride Injection BP

Dextran 70 Injection BP 6% in 5% Glucose Injection BP

Sodium Bicarbonate Injection is not recommended as a diluent.

Ceftazidime Injection may be stored for up to 12 hours below 25°C or 7 days under refrigeration (2-8°C) at concentrations of between 0.05 mg/mL and 0.25 mg/mL in Intraperitoneal Dialysis Fluid (Lactate) BPC 1973.

Ceftazidime Injection has been found compatible for 12 hours below 25°C or 7 days under refrigeration (2-8°C) when admixed at 4 mg/mL with:

Potassium Chloride 10 mEq/L or 40 mEq/L in 0.9% Sodium Chloride Injection BP, or Heparin (10 and 50 units/mL) in 0.9% Sodium Chloride.

Ceftazidime Injection (4 mg/mL) has been found compatible for 24 hours when stored below 25°C or 7 days when refrigerated (2-8°C do not freeze) when admixed with Cloxacillin. Ceftazidime Injection (5 mg/mL) is compatible for 12 hours when stored below 25°C or 7 days when refrigerated (2-8°C do not freeze) when admixed with metronidazole.

Ceftazidime Injection may be reconstituted for intramuscular administration using 0.5% Lignocaine Hydrochloride Injection BP; the resultant solutions may be stored for 6 hours below 25°C or 4 days under refrigeration (2-8°C). Solutions range from light yellow to amber depending on concentration, diluent and storage conditions used. Within the stated recommendations, product potency is not adversely affected by such colour variations.

ROUTE OF ADMINISTRATION: For i.m. or i.v. injection

CONTRAINDICATIONS:

Ceftazidime is contraindicated in persons who have shown hypersensitivity to cephalosporins; and who have experienced: a major allergy to penicillin - anaphylaxis, angioneurotic oedema, urticaria. Lignocaine should not be used as a diluent for i.m. injection in patients hypersensitive to Lignocaine.

WARNINGS & PRECAUTIONS:

Patients with impaired renal functions: Ceftazidime has shown some evidence of renal toxicity in animals. Clinically transient elevations in serum urea and serum creatinine have been shown, almost entirely excreted by glomerular filtration and its half-life is prolonged in renally impaired patients. Dosage adjustment may be required in such patients to avoid clinical consequences of elevated antibiotic levels. It has been reported that neurological sequelae has occurred occasionally when the dose has not been reduced appropriately.

Use in patients with impaired liver functions: When given Ceftazidime transient rises in hepatic enzymes have been noted in some patients hence it is advisable to carefully monitor liver function when any dysfunction exists.

Avoid repeated use of lignocaine diluent for i.m. use in patients with severe liver disease or decreased hepatic blood flow due to the possibility of lignocaine toxicity resulting from decreased metabolism and consequent accumulation.

Prolonged use of ceftazidime may result in the overgrowth of non-susceptible organisms (eg. *Candida enterococci*) which may require interruption of treatment or adoption of appropriate measures.

Use in children: Ceftazidime is effective in the treatment of neonatal infections caused by susceptible organisms.

Warnings: Being a beta-lactam antibiotic a careful inquiry should be made for a history of hypersensitivity reactions to ceftazidime, cephalosporins, penicillins or other drugs before beginning ceftazidime therapy.

Patients with Type I or immediate hypersensitivity reactions to penicillin should be given ceftazidime only with special caution. If an allergic reaction to ceftazidime occurs, discontinue the drug. Serious hypersensitivity reactions may require adrenaline, hydrocortisone, anti-histamine or other emergency measures. Serious and occasionally fatal hypersensitivity reactions (including anaphylactoid and severe cutaneous adverse reactions) have been reported in patients receiving therapy with beta-lactams. Before initiating therapy with Cefatum, careful inquiry should be made concerning previous hypersensitivity reactions to penicillins, cephalosporins, carbapenems or other beta-lactam agents. If an allergic reaction occurs, Cefatum must be discontinued immediately and appropriate alternative therapy should be instituted.

Antibiotic associated pseudomembranous colitis has been reported with ceftazidime. The primary cause appears to be a toxin produced by *Clostridium difficile*. Severity of the colitis ranges from mild to life threatening. Mild cases usually respond to drug discontinuation alone.

In moderate to severe cases appropriate therapy such as oral antibacterial agents effective against *Clostridium difficile* (eg. oral vancomycin) should be considered. Replacement therapy of fluids, electrolytes and protein should be provided when indicated. Drug which delay peristalsis eg. opiates and diphenoxylate with atropine may prolong and/or worsen the condition and should not be used. *Clostridium difficile* infection rarely manifests as diarrhoea in neonates. Peak concentrations of ceftazidime in the CSF are considerably lower than in the plasma. At present its use in the treatment of infections of the CNS eg. meningitis, brain abscess, etc. is not advised.

INTERACTION WITH OTHER MEDICAMENTS:

Chloramphenicol has been shown to be antagonist to beta- lactam antibiotics, including ceftazidime, based on in-vitro studies and time kill curves with enteric gram-negative bacilli. Due to the possibility of antagonism in vivo, particularly when bactericidal activity is desired, this drug combination should be avoided.

Nephrotoxicity has been reported following concomitant use of cephalosporins with aminoglycoside, antibiotics or potent diuretics, such as furosemide.

USE IN PREGNANCY & LACTATION:

Use in pregnancy: Safety of ceftazidime in pregnancy has not been established, therefore it may be administered during known or suspected pregnancy only if in the opinion of the treating physician the expected benefits outweigh the possible risks.

Use in lactation: As ceftazidime is excreted in human breast milk in low concentrations, it is therefore not recommended for nursing mothers unless the benefits to the mother greatly outweighs any risk to the infant.

SIDE EFFECTS:

Adverse reactions are infrequent and include:

Local: phlebitis or thrombophlebitis with i.v. administration; pain and/or inflammation after i.m. injection.

Hypersensitivity: maculopapular or urticarial rash, fever, pruritus, and very rarely angioedema and anaphylaxis (including, bronchospasm and hypotension), Stevens Johnson Syndrome, and toxic epidermal necrolysis.

Gastrointestinal: diarrhoea, nausea, vomiting, abdominal pain, and very rarely oral thrush or colitis. Psuedomembranous colitis has been reported.

Central Nervous System: headache, dizziness, paraesthesia and bad taste.

There have been reports of neurological sequelae including tremor, myoclonia, convulsions, and encephalopathy occurring in patients with renal impairment in whom the dose of ceftazidime has not been appropriately reduced.

Genito-urinary: candidiasis, vaginitis.

Renal: transient elevations of blood urea, serum urea and/or serum creatinine have been observed occasionally.

Hepatic: Elevations in one or more of the hepatic enzymes, SGOT, SGPT, GGT, LDH and alkaline phosphatase may occur.

Haematological: eosinophilia, positive Coomb's test, thrombocytosis; very rarely, transient leucopenia, haemolytic anaemia, neutropenia, thrombocytopenia and lymphocytosis have been seen

Miscellaneous: hot flushes, superficial desquamation around injection site.

SYMPTOMS AND TREATMENT OF OVERDOSE:

Overdosage: Ceftazidime overdosage has occurred in patients with renal failure. Reactions have included seizure activity, encephalopathy, asterixis and neuromuscular excitability. Patients who receive an acute overdosage should be carefully observed and given supportive treatment. In the presence of renal insufficiency, haemodialysis or peritoneal dialysis may aid in the removal of ceftazidime from the body.

INCOMPATIBILITIES:

The admixture of beta-lactam antibiotics (penicillins and cephalosporins) and aminoglycosides may result in substantial mutual inactivation. If they are administered concurrently, they should be administered in separate sites. Do not mix them in the same intravenous bag or bottle.

Vancomycin is physically incompatible with ceftazidime and a precipitate may form, depending on the concentration. Therefore, the intravenous lines should be flushed between the administration of these 2 medications if they are to be given through the same tubing.

STORAGE CONDITIONS:

Store below 30°C, protect from light. Keep out of reach of children. *Jauhkan daripada kanak-kanak.*

SHELF LIFE:

Please refer to outer package.

PACK SIZE:

CEFATUM INJECTION 1 G: Box of 1's x 1g, 5's x 1g and 25's x 1 g in 10 ml vial

CEFATUM INJECTION 2 G: Box of 1's x 2g, 5's x 2g and 25's x 2 g in 20 ml vial

PRODUCT REGISTRATION HOLDER & MANUFACTURER:

DUOPHARMA (M) SDN BHD

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